WORKED SOLUTION

STEP III 1996, Question 7

Pure • Symmetric Functions of Roots

STEP III 1996, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q7

The key idea

Both parts turn on the same fact: any symmetric expression in three numbers can be written in terms of their sum, the sum of their products in pairs, and their product, and these three quantities are exactly the coefficients of the cubic whose roots are the numbers. In (i) this turns a nonlinear system into a cubic equation. In (ii) we never need the roots at all: Heron's formula is symmetric in a, b, c, so the cubic's coefficients are enough.

Solution

Part (i)

First, the sum of squares:

$$\alpha^2 = (x+y+z)^2 = x^2+y^2+z^2+2(xy+yz+zx), \qquad \text{so} \qquad x^2+y^2+z^2 = \alpha^2 - 2\beta.$$

Next we use the factorisation

$$x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2-xy-yz-zx),$$

which you can verify by expanding the right-hand side: the nine cross terms such as $x \cdot xy$ and $y \cdot (-xy)$ cancel in pairs, and the three terms of the form $-xyz$ remain. Substituting,

$$\begin{aligned} x^3+y^3+z^3 &= 3\gamma + \alpha(\alpha^2 - 2\beta - \beta) \\ &= \alpha^3 - 3\alpha\beta + 3\gamma. \end{aligned}$$

So $A = 1$, $B = -3$ and $C = 3\gamma$. (Here C is three times the product, so it depends on γ; A and B are pure numbers.)

Now for the equations. We have $\alpha = 1$, and the sum of squares gives

$$3 = \alpha^2 - 2\beta = 1 - 2\beta \quad \Longrightarrow \quad \beta = -1.$$

The sum of cubes then gives

$$4 = \alpha^3 - 3\alpha\beta + 3\gamma = 1 + 3 + 3\gamma \quad \Longrightarrow \quad \gamma = 0.$$

Ask yourself We know α, β and γ. How do we get back to x, y, z?

The numbers x, y, z have sum α, pairwise-product sum β and product γ exactly when they are the three roots of

$$(t-x)(t-y)(t-z) = t^3 - \alpha t^2 + \beta t - \gamma = 0.$$

Both directions hold: if x, y, z solve the system then they are the roots of this cubic, and conversely the roots of the cubic, in any order, have the right α, β, γ and hence (by the identities above) satisfy all three equations. Here the cubic is

$$t^3 - t^2 - t = t\,(t^2 - t - 1) = 0, \qquad \text{so} \qquad t = 0, \quad t = \tfrac{1}{2}(1+\sqrt{5}), \quad t = \tfrac{1}{2}(1-\sqrt{5}).$$

Therefore the solutions are

$$\{x, y, z\} = \left\{ 0,\ \tfrac{1}{2}(1+\sqrt{5}),\ \tfrac{1}{2}(1-\sqrt{5}) \right\},$$

in any order: six ordered triples (x, y, z) in all, since the three values are different.

Part (ii)

Since a, b, c are the roots of the cubic,

$$x^3 - 16x^2 + 81x - 128 = (x-a)(x-b)(x-c)$$

for every x. Comparing the x^2 coefficients, $a + b + c = 16$, so $s = 8$. Now the trick: put $x = s = 8$ into this identity. The right-hand side becomes $(s-a)(s-b)(s-c)$, the very product in Heron's formula:

$$(s-a)(s-b)(s-c) = 8^3 - 16 \cdot 8^2 + 81 \cdot 8 - 128 = 512 - 1024 + 648 - 128 = 8.$$

Hence

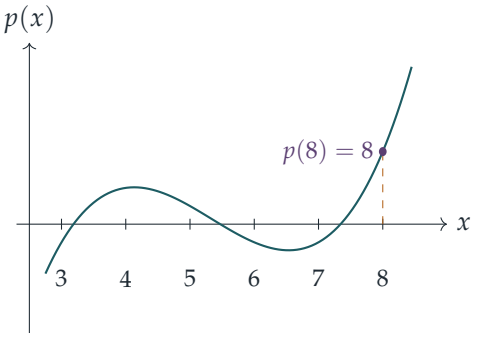
$$\text{area} = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{8 \cdot 8} = 8.$$

Ask yourself Is there really a triangle with these sides?

We should check, because Heron's formula only describes a triangle when a, b, c are positive and each is less than the sum of the other two. Since $a < b + c \iff 2a < a + b + c \iff a < s$, we need all three roots to be real, positive and less than 8. Write $p(x) = x^3 - 16x^2 + 81x - 128$. Then

$$p(3) = -2, \quad p(4) = 4, \quad p(5) = 2, \quad p(6) = -2, \quad p(7) = -2, \quad p(8) = 8,$$

so by the changes of sign there is a root in each of $(3,4)$, $(5,6)$ and $(7,8)$. A cubic has at most three roots, so these are all of them, and every root lies strictly between 0 and 8. The triangle exists, and its area is 8.



Checks

- In (i), try $x = 1$, $y = 2$, $z = 3$: $\alpha = 6$, $\beta = 11$, $\gamma = 6$ and $216 - 198 + 18 = 36 = 1 + 8 + 27$.
- With $x = 0$, $y = \frac{1}{2}(1 + \sqrt{5}) \approx 1.618$, $z = \frac{1}{2}(1 - \sqrt{5}) \approx -0.618$: the sum is 1, and since $y^2 = y + 1$ and $z^2 = z + 1$, the squares sum to $(y + z) + 2 = 3$ and the cubes to $y^2 + z^2 + (y + z) = 3 + 1 = 4$.
- Numerically the roots in (ii) are $a \approx 3.186$, $b \approx 5.471$, $c \approx 7.343$. Heron's formula with these values gives 8.000, and so does $\frac{1}{2}ab\sin C$ with C from the cosine rule.

Common slips

- Giving one triple instead of all six orderings, or forgetting that $t = 0$ is a root of $t^3 - t^2 - t = 0$.
- Getting a sign wrong in Vieta's formulas: for $t^3 - \alpha t^2 + \beta t - \gamma$ the signs alternate.
- Taking the area without checking that the roots really form a triangle.

Beyond STEP

The fundamental theorem of symmetric polynomials says that every polynomial in x, y, z that is unchanged by permuting them is a polynomial in the elementary symmetric functions α, β, γ. Newton's identities give the power sums $x^k + y^k + z^k$ recursively. These ideas lead into Galois theory, which explains why there are formulas for the roots of cubics and quartics but none for general quintics. Allenby's *Rings, Fields and Groups* is a good first step in that direction.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.